Dhad Readability Classifier

Multi-label difficulty scoring, built with an agent-assisted, evaluation-first workflow, and tested on words the model has never seen.

Bisharah Estephan · 29 October 2026

0.805	+0.65	21	13
val macro F1 floor 0.524	encoder response to unseen technical words	commits in 92 minutes	AI agents, each with a narrow brief

AT A GLANCE

A model that reads difficulty three ways

Given a short English passage, the model returns three independent probabilities: **lexical load**, **syntactic complexity** and **conceptual abstraction**.

The delivered TF-IDF + logistic regression model scores **macro F1 0.805** (95% CI 0.75–0.86) on the validation set, against 0.524 for always predicting "present". Validation data was opened **exactly once**, after the model and thresholds were frozen under the git tag `pre-val`.

> **The central finding: unseen vocabulary.** In this data each label hangs on a handful of slot words, so ordinary cross-validation rewards memorisation. Holding whole words out drops the baseline from 0.828 to 0.741; a fine-tuned DeBERTa-v3 encoder reaches 0.772. Shown words that never occur in training, the baseline barely reacts (+0.10, +0.22), while the encoder responds almost as strongly as to familiar words (+0.65, +0.74).

01 · THE PROJECT

What was asked, what was delivered

THE BRIEF ASKED FOR	DELIVERED
Training code that reads train.csv and writes a checkpoint	✓ `uv run dhad-train`: CV, threshold tuning, refit, checkpoint and run manifest
Three probabilities per passage (multi-label)	✓ `predict_proba` returns an (n, 3) array in [0, 1]
Evaluation on val.csv: per-dimension and macro F1	✓ `uv run dhad-evaluate`: F1 with bootstrap CIs, plus AUC, AP, Brier and a shortcut check
README of one page or less, 2–3 design choices	✓ Three design choices, results and limitations
Local compute; no rules written from val	✓ Mac + personal RTX 5090; val read by one module, once

02 · HOW I WORKED

Evaluation-first, agent-assisted

I set the rules first, then used AI agents for breadth: research, review, parallel implementation and QC. Every decision that touches evaluation stayed with me.

▼ validation firewall: nothing to the left ever saw val.csv

Discover	Plan	Review	Build baseline	Freeze	val.csv	Encoder	Behaviour tests	Final QC
4 research agents	committed first	3 review agents	+ 2 build agents	tag pre-val	read once	RTX 5090	seen vs unseen	4 read-only agents

Thirteen agents, each with a narrow brief

Research · 4

Modelling, evaluation methodology, engineering stack, and a train-only analysis of the data.

Plan review · 3

Model choice and licences checked on Hugging Face, a data scientist and readability expert, and a grader's view.

Build · 2

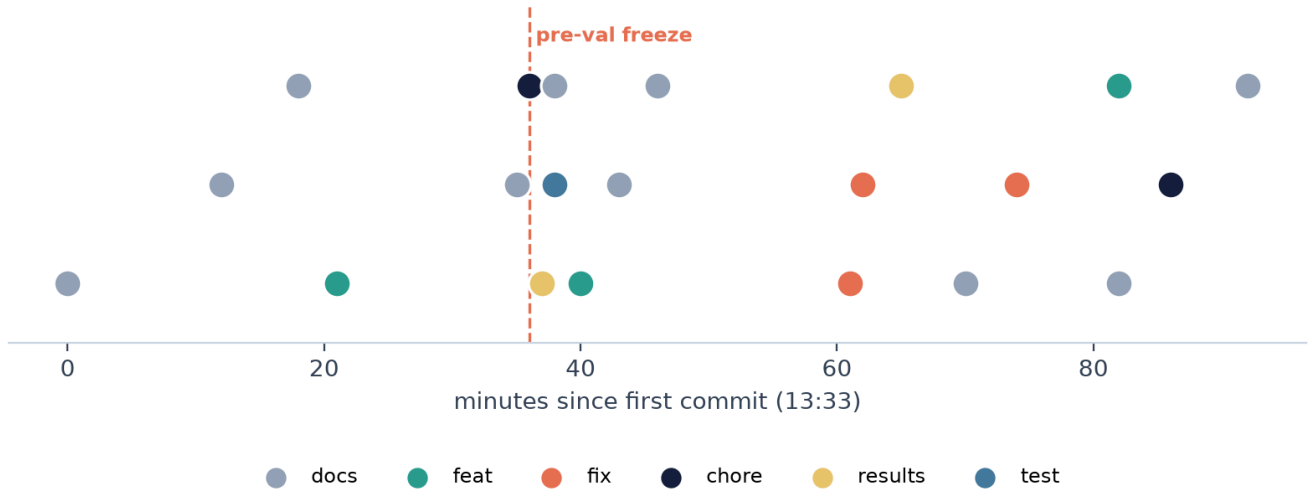
Unit tests and the analysis script, written in parallel on files that did not overlap.

Final QC · 4, read-only

Numbers against result files, code correctness, privacy and git hygiene, and the brief's deliverables.

Agent output was checked, not trusted. The reviews reordered the plan (baseline deliverables first, encoder time-boxed). One agent's claim of swapped figures was checked and rejected. QC findings were applied only when verified, and correct but untestable code issues were documented rather than patched blind.

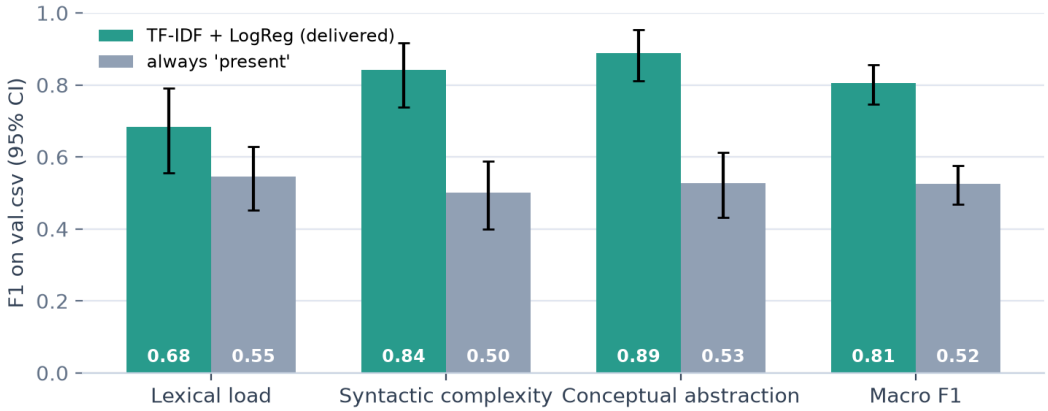
Delivery timeline



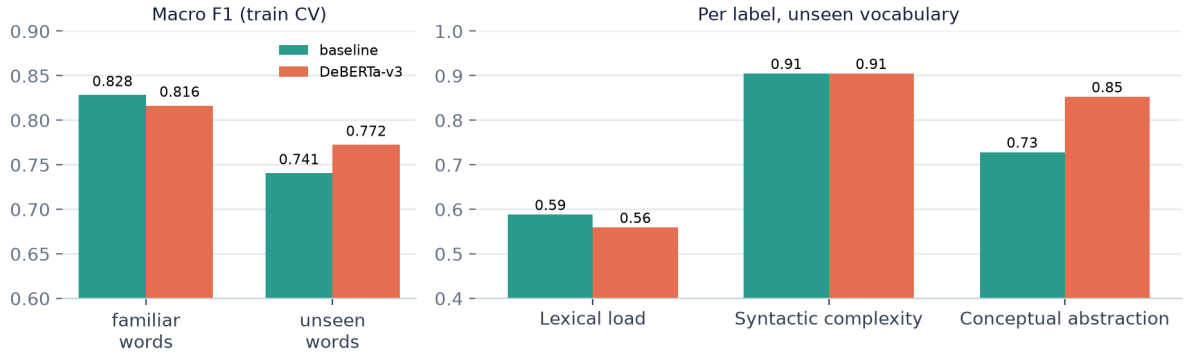
21 commits in 92 minutes, coloured by conventional-commit type. The freeze precedes the single validation run.

03 · RESULTS

Well above the floor, honest about the noise



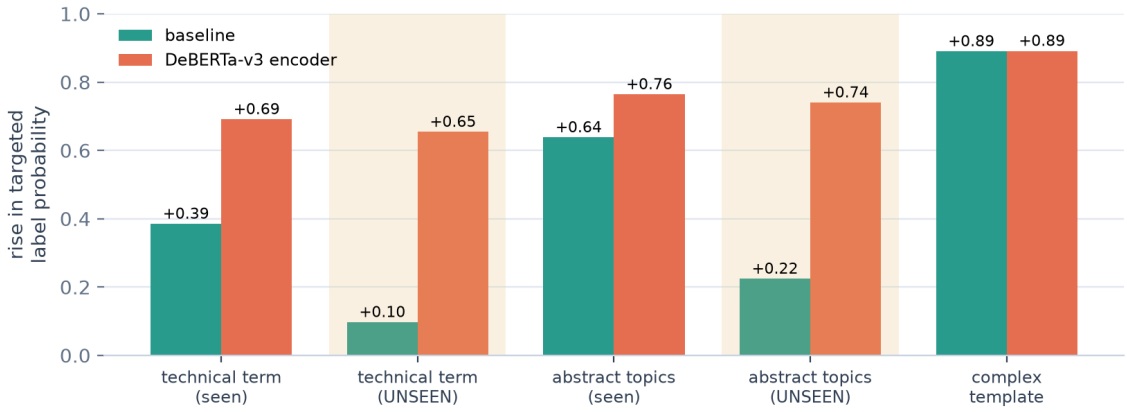
Validation F1 with 95% bootstrap CIs (2,000 resamples), computed once, after the freeze. With 120 rows, gaps under about 0.05 are noise.



Train cross-validation. On unseen words the encoder holds up; the gain is in abstraction, while lexical load does not improve.

What did each model actually learn?

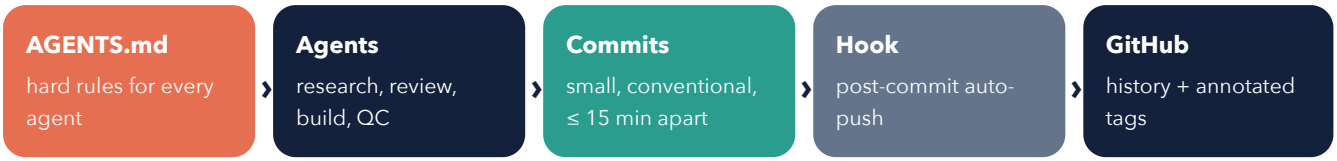
F1 cannot show *why* a model is right. Behavioural tests change one factor at a time in 200 minimal pairs built from the training templates, using words seen in training and words verified never to occur in it (*isotope, jurisprudence, morality, virtue*). Validation data is not used.



Rise in the targeted label's probability. Highlighted: unseen words, where only the encoder generalises. Off-target shifts stay at 0.04 or less, except a shared shortcut: the complex template alone raises lexical load.

04 · GOVERNANCE

How the GitHub repository keeps the work honest



MECHANISM	WHAT IT GUARANTEES
AGENTS.md (checked in)	val never used for training, tuning or selection; every number from a real run; no secrets or large files; no history rewriting.
`pre-val` annotated tag	Model and thresholds frozen in results/selection.json before val.csv was read; the val results commit comes after it.
Single val reader	Only `dhad-evaluate` reads val.csv; tests and behavioural tests never do.
post-commit hook	Every commit is pushed immediately; failures are logged to .git/push.log.
Conventional commits	feat / fix / test / docs / results / chore, with Co-Authored-By trailers that make agent involvement visible.
Run manifests	Each run records its git SHA, the SHA-256 of its data and the chosen settings.
Plan and progress in docs/	Research, plan, review revisions and a story-point tracker are committed, so the process can be audited.

05 · TOOLS

Claude Code + parallel sub-agents · uv · uv.lock · Python 3.12+ · scikit-learn · PyTorch 2.15 · transformers 5.21 · DeBERTa-v3-base (MIT) · iterative-stratification · pytest · 20 tests · ruff · matplotlib · git + GitHub · personal RTX 5090

Honest limitations

- Syntactic complexity is learned from the template (F1 0.00 within the simple template, 0.94 within the complex one).
- Synthetic, noisy labels cap the reachable F1.
- 120 validation rows: gaps under about 0.05 F1 are noise.
- The encoder came after the freeze: exploratory, no val score. Two bugs found on the way were fixed and documented.

Next: transfer to Arabic

- Arabic encoder (CAMeLBERT-MSA, AraBERTv2); held-out-vocabulary CV becomes held-out-root CV.
- Lexical load from root and pattern rarity; syntax from clitic and iḍāfa depth.
- Human-labelled ordinal difficulty levels instead of binary flags.